# 17.01 Transmission modes

For communication over an internetwork there are two possible approaches: circuit switching or packet switching.

## Circuit switching

Circuit switching is the method used in the traditional telephone system. Because the Public Switched Telephone Networks (PTSNs) have now largely converted to digital technology, the same method can be provided for data transfer that was traditionally used for voice communication. In Chapter 2 (Section 2.06) the concept of support for Internet usage by a PTSN was introduced. Typically, this is provided in a leased line service. The concept is illustrated in Figure 17.01, which shows end-systems connected to local exchanges which have a switching function and which are connected via a number of intermediate nodes with a switching function.

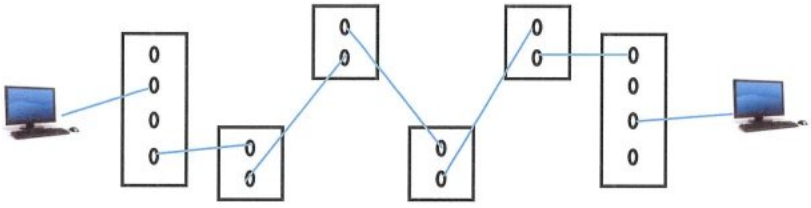


Figure 17.01 Circuit-switched data transmission



For data transfer to take place, the following has to happen.

1. The sender provides the identity of the intended receiver.
2. The system checks whether or not the receiver is ready to accept data.
3. If the receiver is available, a sequence of links is established across the network.
4. The data is transferred.
5. The links are removed.

It is not necessary for this discussion to define what could constitute a node in a circuit-switched network. The links that are provided between the nodes are dedicated channels in shared transmission media that guarantee unimpeded transmission. When a telephone call has finished there is a definite end to the call with removal of the links. However, for a leased-line data connection there might be a permanent circuit established.

## Packet switching

The packet-switching method allows data transmission without a circuit being established. Data cannot be sent in a continuous stream. Instead data is packaged in portions inside packets. A packet consists of a header which contains instructions for delivery plus the data body. The method is similar to that used by the postal service but rather more complex! The network schematic shown in Figure 17.01 is still appropriate to describe packet switching except that the links used are not defined at the time a packet is transmitted by the sender. Furthermore, the nodes will have an extended functionality compared to that required in a circuit-switched transmission. How a router acts as a node and supports packet switching is discussed in Section 17.04.

When packet switching is used, there are two ways that the network can provide a service: connectionless service or connection-oriented service. If a connectionless service is provided, a packet is dispatched with no knowledge of whether or not the receiver is ready to accept the packet, and has no way of finding out if the transmission has

succeeded. In a connection-oriented service the first packet sent includes a request for an acknowledgement. If the acknowledgement is received, the sender transmits further packets. If no acknowledgement is received, the sender tries again with the first packet.

## 17.02 Protocols

The basic definition of a **protocol** is simple – it is a set of rules. So, what do these rules relate to? Before answering, it should be understood that we often talk about 'a protocol' when we are referring to a protocol suite which contains more than one individual protocol. The complexity of networking requires many individual protocols. A further complication is that there might be a number of different versions of a protocol. Finally, there are often protocols that can be used to complement the use of another protocol.

KEY TERM

**Protocol:** a set of rules for data transmission which are agreed by sender and receiver

The set of rules that constitute a protocol must be agreed between the sender and the receiver for any communication transmitted over a network. At the simplest level, a protocol could define that a positive voltage represents a bit with value 1. A protocol might define a transmission speed that a sender must not exceed. Many of the rules relate to the format of a message or of a component of a message. For example, a definition of the format of the first 40 bytes in a packet.

## 17.03 A protocol stack



For a protocol suite the protocols can be viewed as layers within a protocol stack. There are a number of aspects relating to this concept.

- Each layer can only accept input from the next higher layer or the next lower layer.
- There is a defined interface between adjacent layers which constitutes the only interaction allowed between layers.
- A layer is serviced by the actions of lower layers.
- With the possible exception of the lowest layer the functioning of a layer is created by installed software.
- A layer may comprise sub-layers.
- Any user interaction will take place using protocols associated with the highest level layer in the stack.
- Any direct access to hardware is confined to the lowest layer in the stack.

## 17.04 The TCP/IP protocol suite

TCP/IP is the protocol suite underpinning Internet usage. TCP/IP can be explained on the basis of the network model shown in Figure 17.02. It can be seen that TCP/IP only occupies the top three layers of this model.

TIP

Unfortunately there are different names used for two of the layers. Figure 17.02 shows the Network layer and the Data link layer. In some sources these will be named as the Internet layer and the Link layer.